Inro to GeN-Foam: theory

C. Fiorina

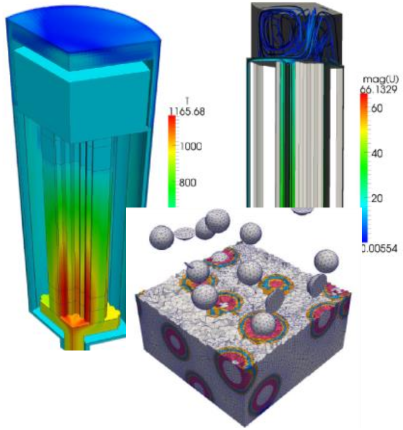
OpenFOAM for multi-physics (in nuclear)

Carlo Fiorina

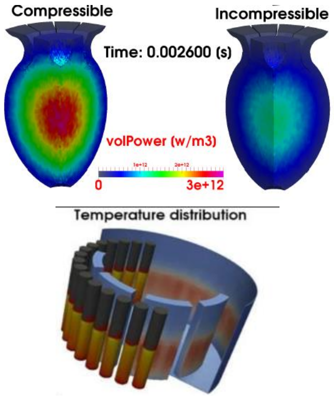
2000-2010
First activities

2010-2015
First widespread use

2015-2022
First coordinated and persistent developments

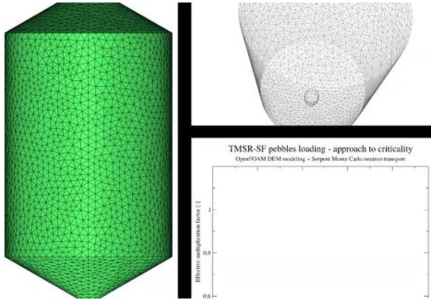


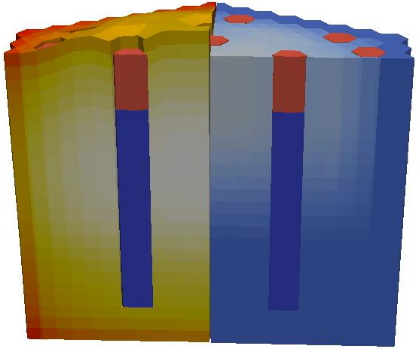
PBMRs and HTRs



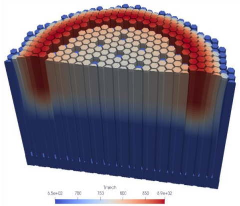
MSRs

FHR

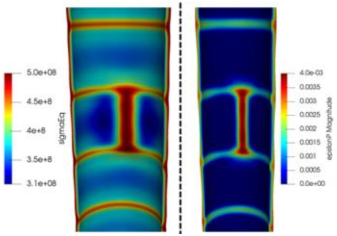




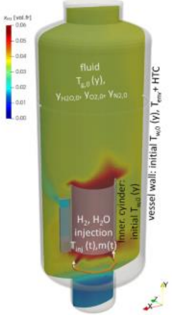
SFRs



GeN-Foam



OFFBEAT



containmentFoam

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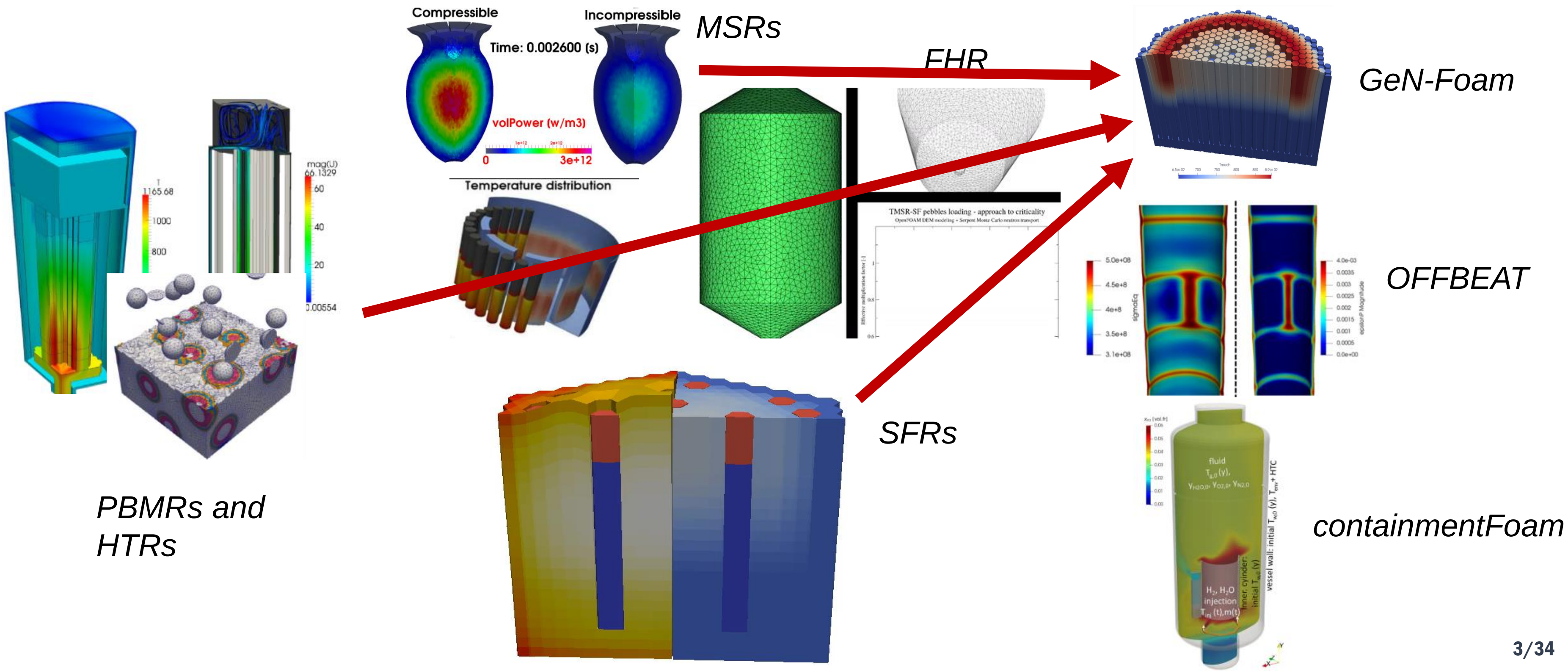

GeN-Foam: Generalized Nuclear Field operation and manipulation

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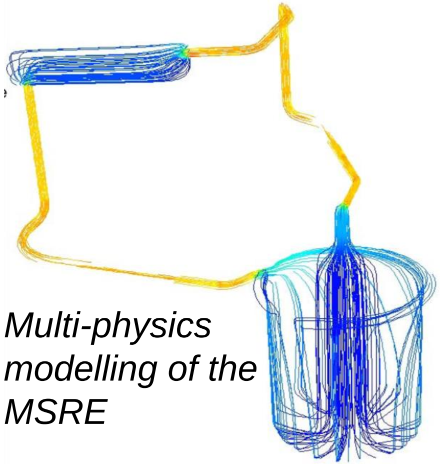
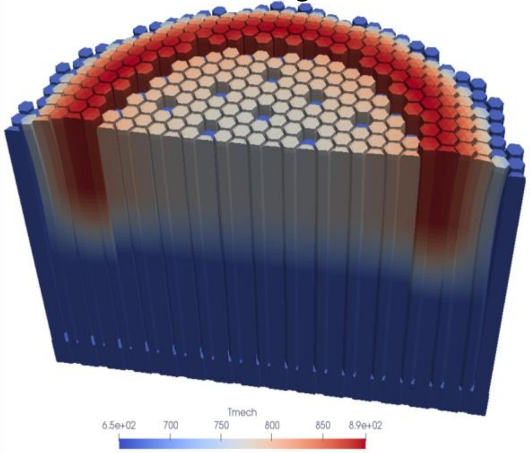
- **Since 2014, EPFL + PSI + growing contributions from various institutions**
- **Complement legacy codes** with more flexibility, mainly targeted **to advanced concepts**
- Distributed to 20+ institutions. Now freely available from GitLab and IAEA website
- Training via the IAEA (webinar, lecture, upcoming series of lectures), workshop at conferences
- Employed in several Euratom proposals, publications from UCam, PoliMi, TAMU, NCSU, UCB

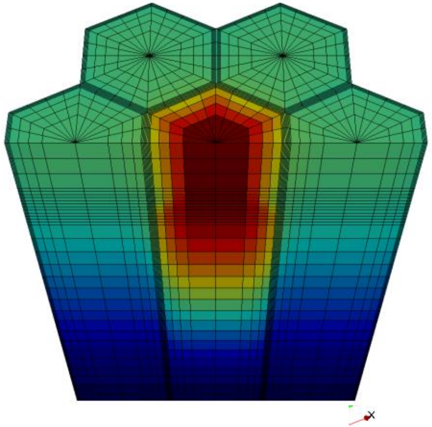


Multi-physics modelling of the MSRE

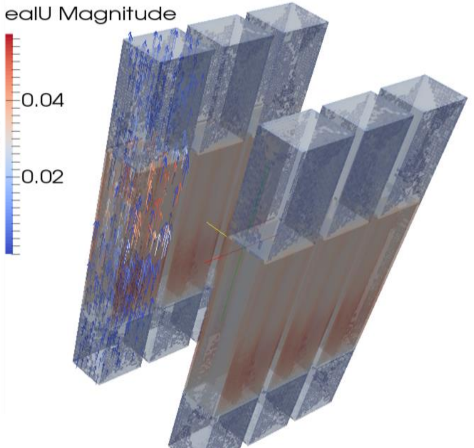
Core flowering in a SFR



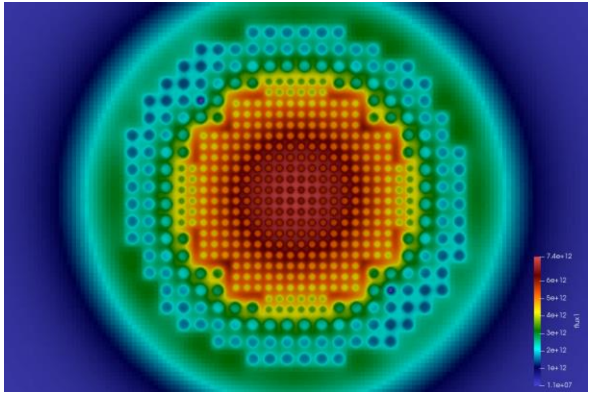
Assembly windows in a SFR

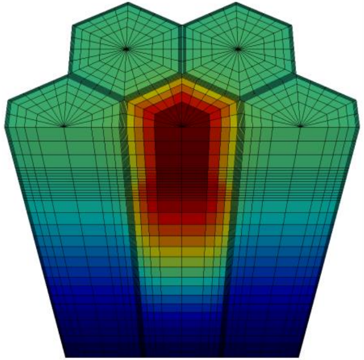


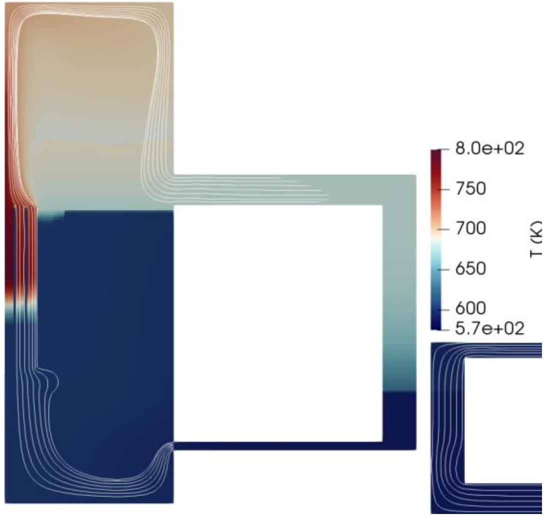
The Argonaut reactor



GeN-Foam: which physics?

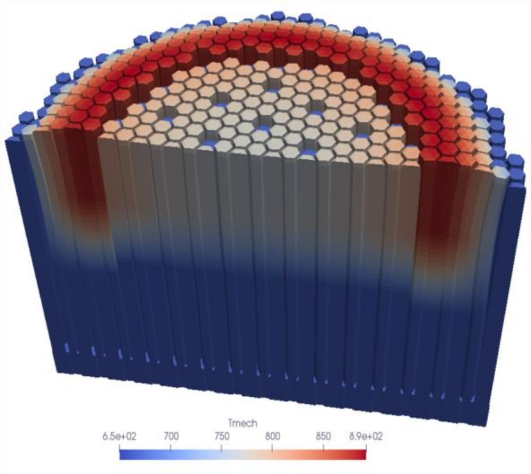






<u>*Neutronics*</u>
- *Diffusion*
- *SP3*
- *SN*
- *Point-kinetics*
- *Precursor transport*

<u>*Thermal-hydraulics:*</u>
- *RANS CFD + porous-mediur*
- *One and two phase*
- *Two phase models for sodium and water (not fully validated)*



<u>*Thermal-mechanics*</u>
- *Linear elasticity*
- *BC for multi-material and contact*

Porous-medium thermal-hydraulics: volume averaging

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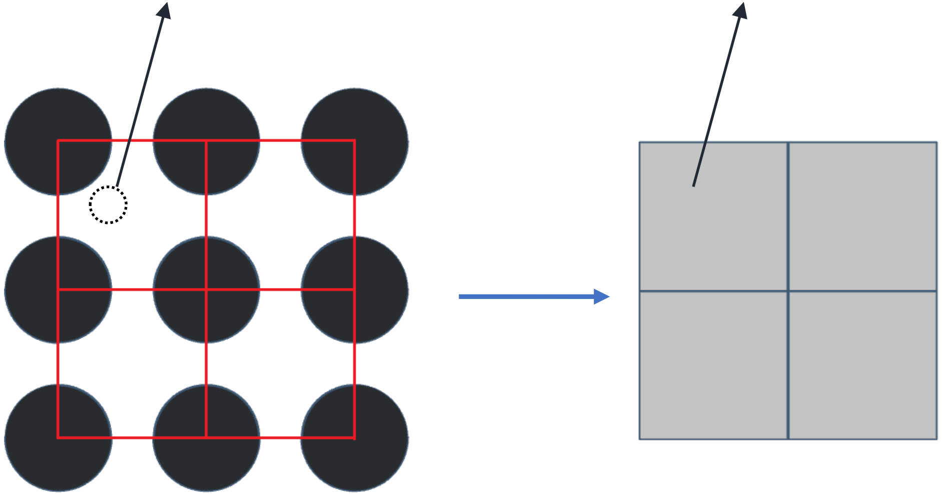
$$\frac{\partial}{\partial t}\rho_i^* + \boldsymbol{\nabla}\cdot\left(\mathbf{u}_i^*\rho_i^*\right) = 0$$

$$\langle \phi_i^* \rangle_I = \frac{1}{V_i}\int_{\Omega_i} \phi_i^* \, dV$$

$$\langle \phi_i^* \rangle = \frac{1}{V}\int_{\Omega_i} \phi_i^* \, dV$$

$$\frac{\partial}{\partial t}\left(\alpha_i\rho_i\right) + \boldsymbol{\nabla}\cdot\left(\alpha_i\mathbf{u}_i\rho_i\right) = \sum_{j\neq i}\frac{1}{V}\int_{\partial\Omega_{ij}} \rho_i^*\left(\mathbf{u}_\partial^* - \mathbf{u}_i^*\right)\cdot\mathbf{n}\, dS$$



Volume averaging results in:

- additional **variables** (phase fraction, tortuosity, etc.);
- additional terms that require **experimentally-informed closure**;

Governing equations

The coarse-mesh governing equations (Navier-Stokes and enthalpy) are:

$$\frac{\partial}{\partial t}\left(\alpha_i\rho_i\right) + \boldsymbol{\nabla}\cdot\left(\alpha_i\mathbf{u}_i\rho_i\right) = -\Gamma_{i\to j}$$

$$\frac{\partial}{\partial t}\left(\alpha_i\rho_i\mathbf{u}_i\right) + \boldsymbol{\nabla}\cdot\left(\alpha_i\rho_i\mathbf{u}_i\otimes\mathbf{u}_i\right) =$$

$$-\alpha_i\boldsymbol{\nabla}p + \boldsymbol{\nabla}\cdot\left(\alpha_i\boldsymbol{\sigma}_{d,i}\right) + \alpha_i\rho_i\mathbf{g} - \mathbf{S}_{\mathbf{u},i\to j}$$

$$\frac{\partial}{\partial t}\left(\alpha_i\rho_i h_i\right) + \boldsymbol{\nabla}\cdot\left(\alpha_i\mathbf{u}_i\rho_i h_i\right) =$$

$$\boldsymbol{\nabla}\cdot\left(\alpha_i\kappa_i\boldsymbol{T}_i\cdot\boldsymbol{\nabla}T_i\right) + \alpha_i\frac{\partial}{\partial t}p + \alpha_i\rho_i\mathbf{u}_i\cdot\mathbf{g} + \alpha_i q_{int,i} - S_{h,i\to j}$$

These reduce to traditional CFD approaches in clear fluid regions and a system-code-like approach in 1-D regions (multiple scales).

GeN-Foam: porous-medium thermal-hydraulics

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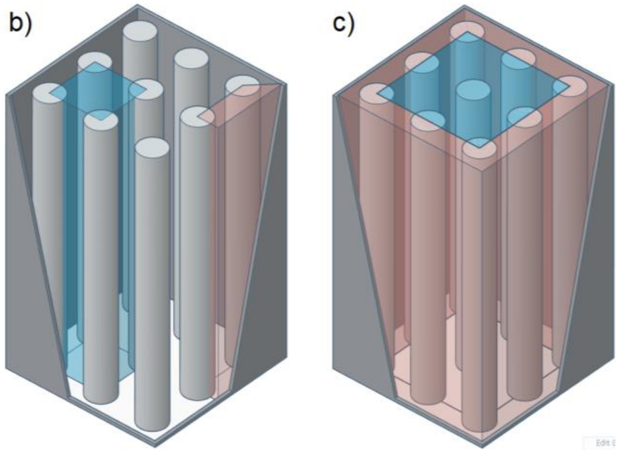
- Porous-medium
  - ✓ 3-D version of a system code, or ore general version of a sub-channel code
  - ✓ Allows for both coarse-mesh (sub-channel like) and fine-mesh (RANS CFD) solutions on the same mesh



$$\boldsymbol{\nabla} \cdot \boldsymbol{u} = 0$$

Volume fraction occupied by the fluid

Pressure drops correlations

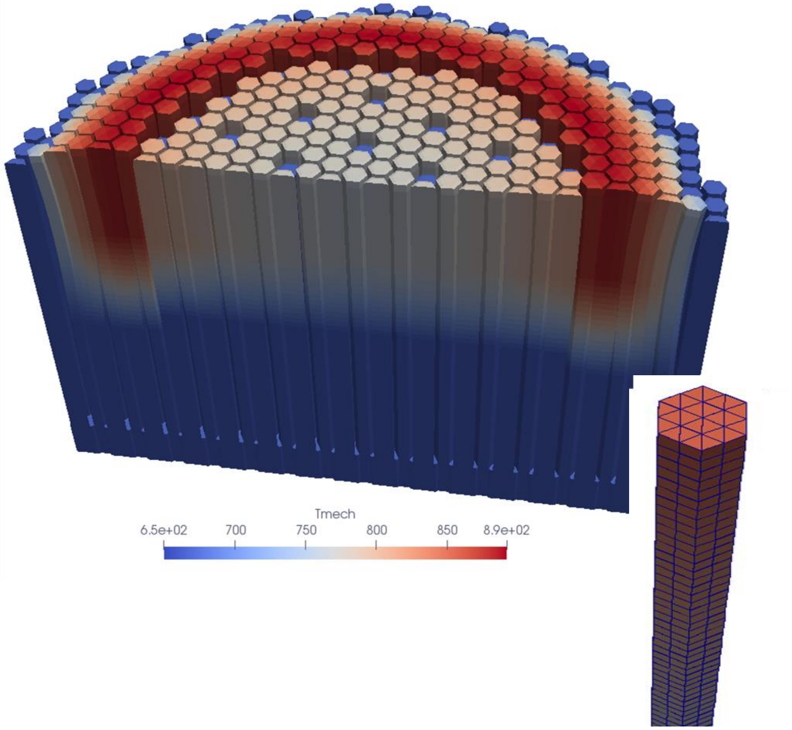
$$\frac{\partial(\chi\rho\boldsymbol{u})}{\partial t} + \boldsymbol{\nabla} \cdot (\chi\rho\boldsymbol{u} \otimes \boldsymbol{u}) = \boldsymbol{\nabla} \cdot (\mu_t \boldsymbol{\nabla}\boldsymbol{u}) - \boldsymbol{\nabla}(\chi p) + \chi\boldsymbol{F}_{\boldsymbol{g}} + \chi\boldsymbol{F}_{\boldsymbol{ss}}$$

Nusselt number correlations

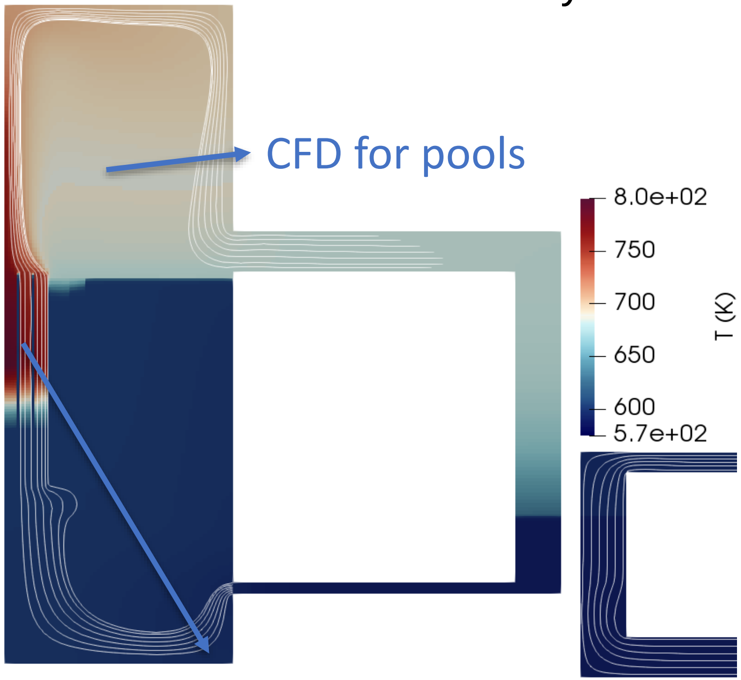
$$\frac{\partial(\chi\rho e)}{\partial t} + \boldsymbol{\nabla} \cdot \left(\chi\rho\boldsymbol{u}\left(e + \frac{p}{\rho}\right)\right) = \boldsymbol{\nabla} \cdot (\chi k_t \boldsymbol{\nabla} T) + \boldsymbol{F}_{\boldsymbol{ss}} \cdot \boldsymbol{u} + \chi\dot{Q}$$

GeN-Foam: porous-medium thermal-hydraulics

3-D coarse mesh simulation of a SFR core

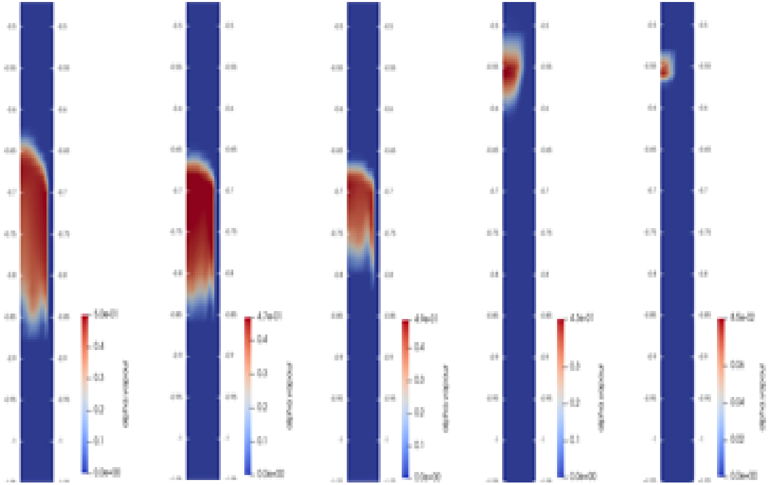


2-D combined coarse/fine mesh simulation of the Fast Flux Test Facility



GeN-Foam: porous-medium thermal-hydraulics

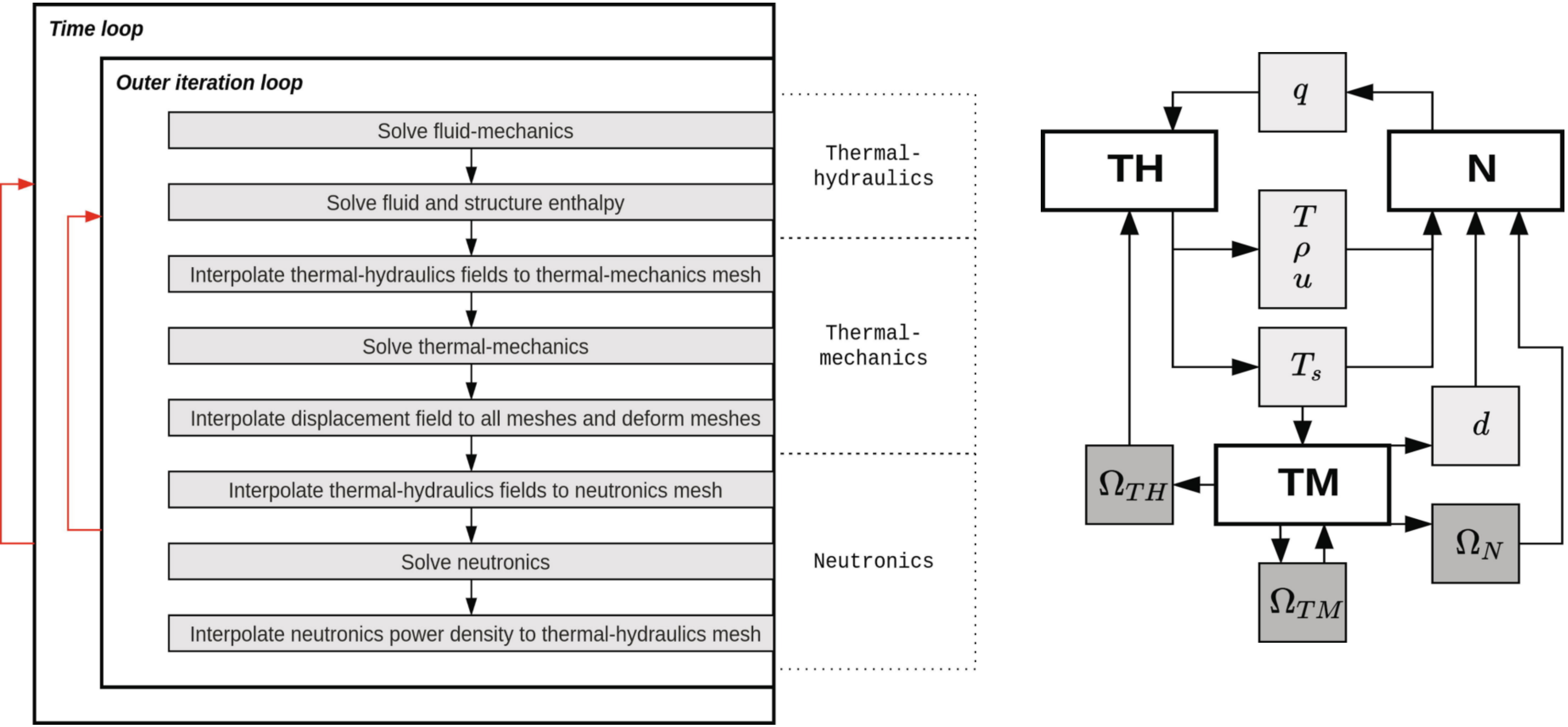
- Same approach as for single-phase thermal-hydraulics (porous-medium with sub-scale structure)
- Beyond the scope of this workshop. Further info in the EPFL PhD thesis of Stefan Radman



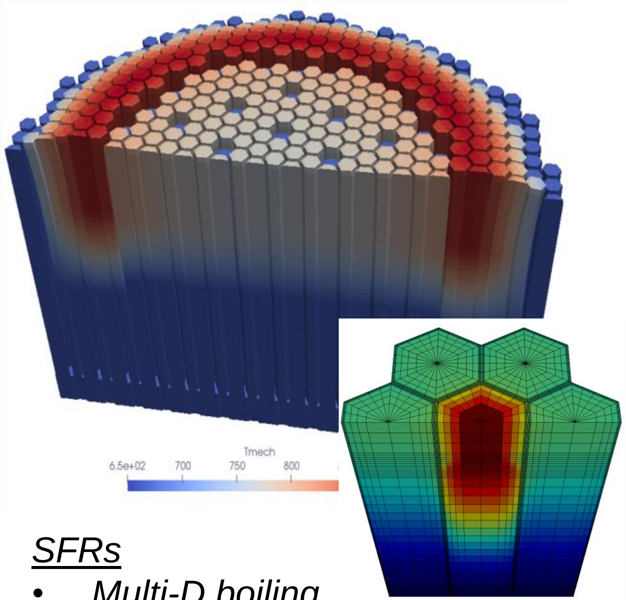
2-D coarse mesh simulation of a SFR assembly with windows

GeN-Foam: coupling

- Semi-implicit (extended PIMPLE loop)

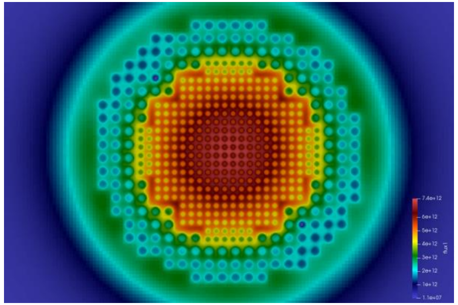


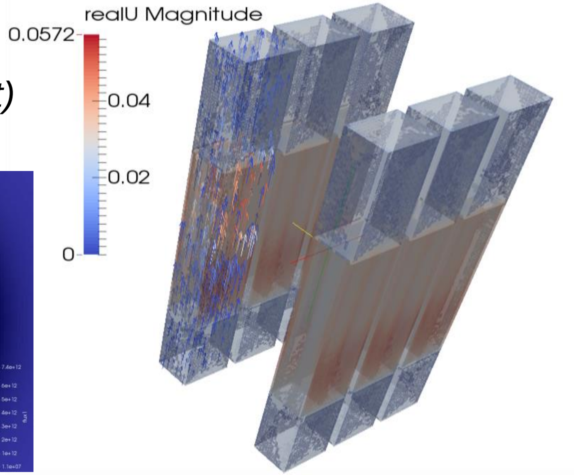
GeN-Foam: which reactors?

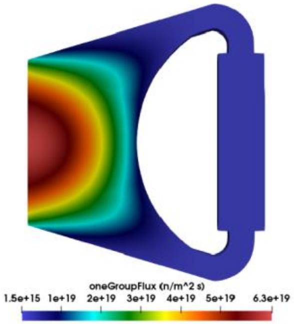


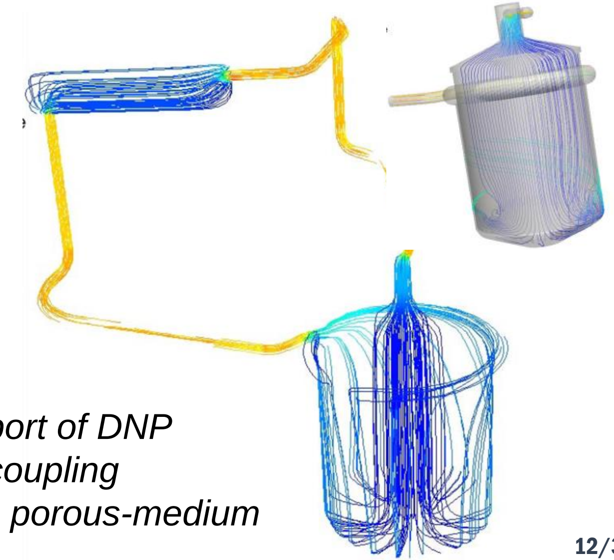
SFRs
- Multi-D boiling
- Coupling of pools and core
- Direct simulation of deformations

Experimental reactors
(ex, CROCUS, Argounaut)
- Geometric flexibility









MSRs
- Transport of DNP
- Tight coupling
- CFD + porous-medium

GeN-Foam: which reactors?

- **LWRs**
  - ✓ TRACE boiling models implemented and tested
  - ✓ Under validation (PSBT)
- **HTRs and FHRs**
  - ✓ Only needs sub-scale model for temperature in pebbles
- **Micro reactors**
  - ✓ Mainly needs modelling skills
- **Heat pipes**
  - ✓ Under development
- …

GeN-Foam: V&V status

Fiorina

Brief description	Neutronics	Thermal-hydraulics	Thermal-mechanics	Coupling
Comparison against PARCS for a PWR a mini-core [15]	x (SP3)			
Comparison against Serpent for the CROCUS reactor [15]	x (SP3)			
Comparison against Serpent for the ESFR [17]	x (Diffusion)			
Comparison against Serpent for a PWR mini-core [17]	x (Diffusion)			
Comparison against various codes for the ESFR-SMART design [21]	x (Diffusion)			
Verification against analytic solutions for a simplified MSR [22]	x (Diffusion)	x (1 phase)		x
Verification against the CNRS MSR benchmark [23]	x (Diffusion)	x (1 phase)		x
Comparison against TRACE for the ESFR core [3,18]	x (Diffusion)	x (1 phase)	x	x
Verification using the method of manufactured solutions [6]		x (1-2 phases)		
Validation against the Godiva IV experiment [16]	x (SN)			
Validation against the FFTF LOFWOS Test 13 [4]	x (pk)	x (1 phase)		
Validation against the KNS-3-L22 experiment on sodium boiling [4]		x (1-2 phases)		
Validation against the ISPRA experiment on sodium boiling [4]		x (1-2 phases)		
Validation against the NEA PSBT benchmark on water boiling		x (1-2 phases)		
Validation against CROCUS measurements [19]	x (Diff, SP3, SN)			

GeN-Foam: Usability

- Complex solver (multi-physics, general finite-volume methodologies on unstructured meshes, linux)
  - ✓ A background on CFD calculations has been observed to greatly reduce the initial barrier
  - ✓ Familiarity with OpenFOAM is necessary

- Still limited documentation
  - ✓ Users must be familiar with what they are modelling

- Flexible solver
  - ✓ Unstructured meshes, several existing sub-solvers, possibility of tailoring

- Particularly suitable for PhD students and researchers that wish to experiment on methods, address particularly complex problems, or investigate non-traditional reactors

- An expanded documentation and set of tutorials have recently made it possible to use GeN-Foam in the frame of shorter projects such as Master Thesis, as well as a tool for education and training.


Thank you for your attention !